

Brain Tumor Detection Using YOLOv8

Summer of Code Project Report

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Project: AI for Healthcare

1. Introduction

Brain tumors are abnormal growths of cells inside the brain that can significantly affect a person's health. Early diagnosis greatly improves treatment planning and patient outcomes. MRI (Magnetic Resonance Imaging) is one of the most commonly used imaging techniques for detecting brain tumors because it provides clear images of soft tissues.

In recent years, deep learning has shown remarkable success in medical image analysis. Among different object detection algorithms, YOLO (You Only Look Once) has become popular because of its speed and accuracy.

In this project, YOLOv8 was used to automatically detect brain tumors in MRI scans by locating the tumor using bounding boxes.

2. Objectives

The objectives of this project were:

- Detect brain tumors from MRI images automatically.
 - Localize tumors using bounding boxes.
 - Train a YOLOv8 object detection model.
 - Evaluate the model using standard performance metrics.
 - Analyze prediction accuracy on unseen MRI images.
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3. Dataset

The dataset used in this project was obtained from **Roboflow Universe**.

Dataset Details

- Dataset: Brain MRI Tumor Detection
- Image Type: MRI scans

- Total Images: Approximately **1229**
- Number of Classes: **1 (Tumor)**

Each image contains manually annotated bounding boxes indicating the tumor location.

The dataset was divided into:

- Training Set
- Validation Set
- Testing Set

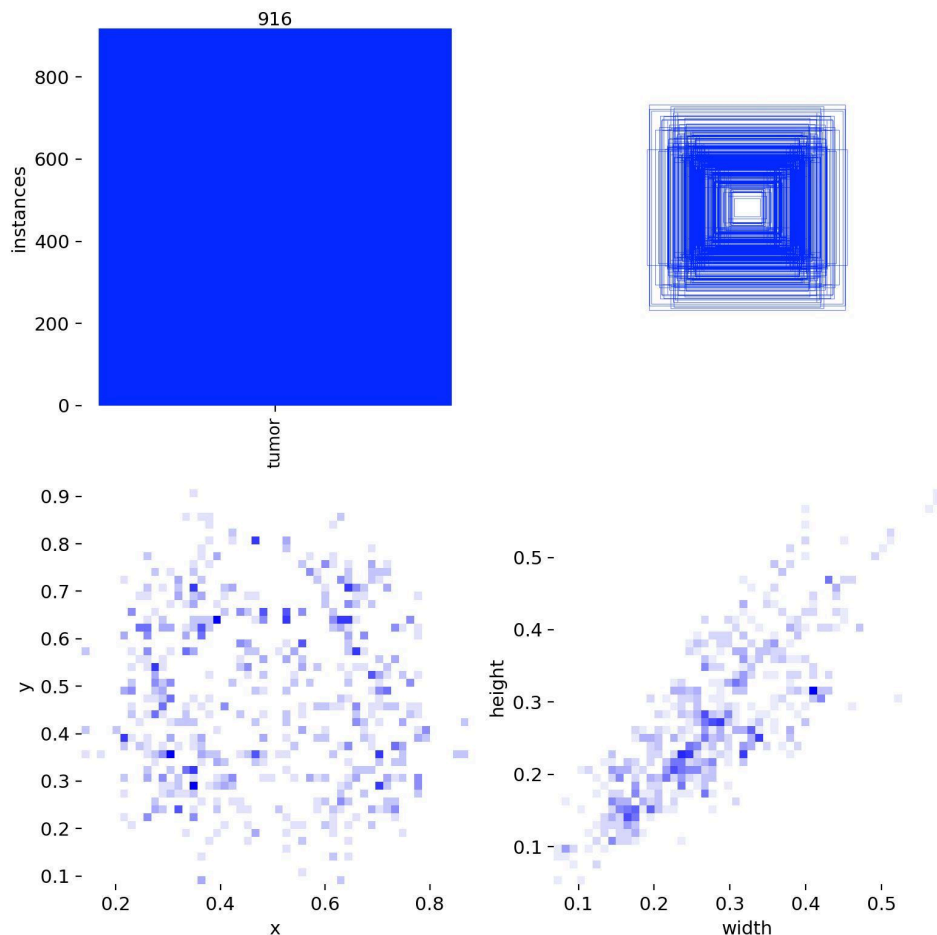


Figure 1 shows the distribution and annotation statistics of the dataset.

4. Methodology

The project was implemented using the YOLOv8 object detection model provided by the Ultralytics library. The overall workflow is shown below:

1. Download the MRI dataset from Roboflow.
2. Prepare the dataset in YOLO format.
3. Train the YOLOv8 model using the training images.
4. Validate the model after each epoch.
5. Evaluate the trained model using standard metrics.
6. Perform predictions on unseen MRI images.

YOLOv8 processes an image in a single forward pass and predicts the location of tumors using bounding boxes along with confidence scores. This makes the model both fast and accurate for medical image detection.

5. Training Configuration

The model was trained using the following configuration:

Parameter	Value
Model	YOLOv8n
Epochs	50
Image Size	640 × 640
Batch Size	16
Optimizer	Default (Ultralytics)
Classes	1 (Tumor)

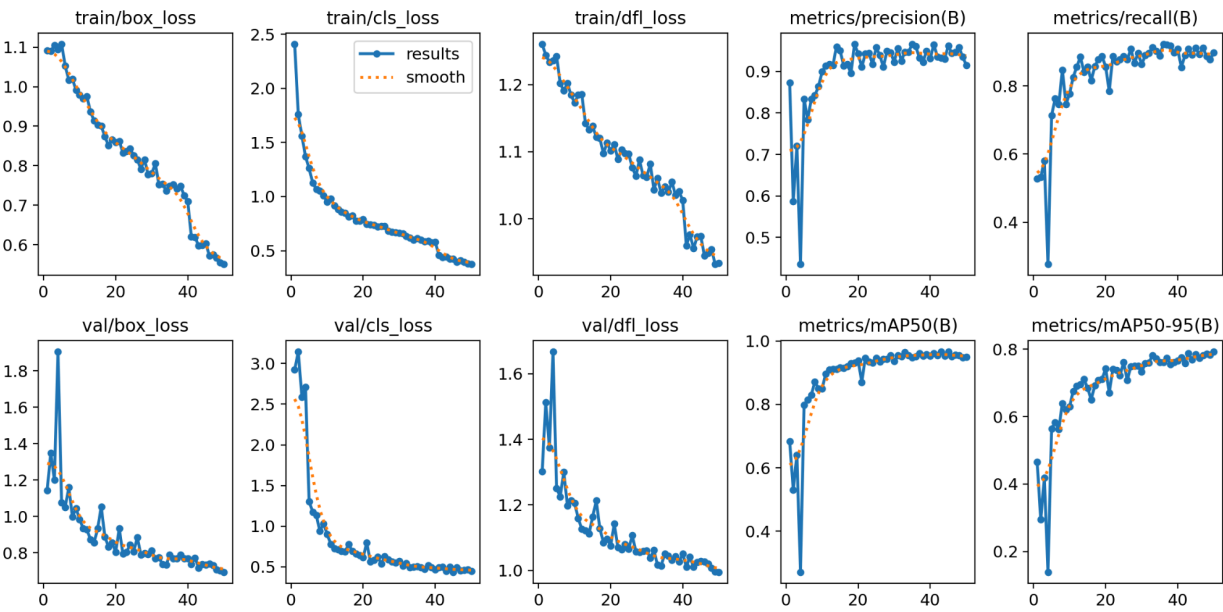
The model was trained on annotated MRI images, where each tumor region was marked using bounding boxes.

6. Results and Discussion

The trained model showed good convergence during training and achieved high detection accuracy on the validation dataset.

6.1 Training Performance

The training and validation losses consistently decreased over the training epochs, indicating that the model successfully learned meaningful features from the MRI images. At the same time, precision, recall, and mAP values steadily improved, demonstrating stable learning without significant overfitting.



6.2 Performance Metrics

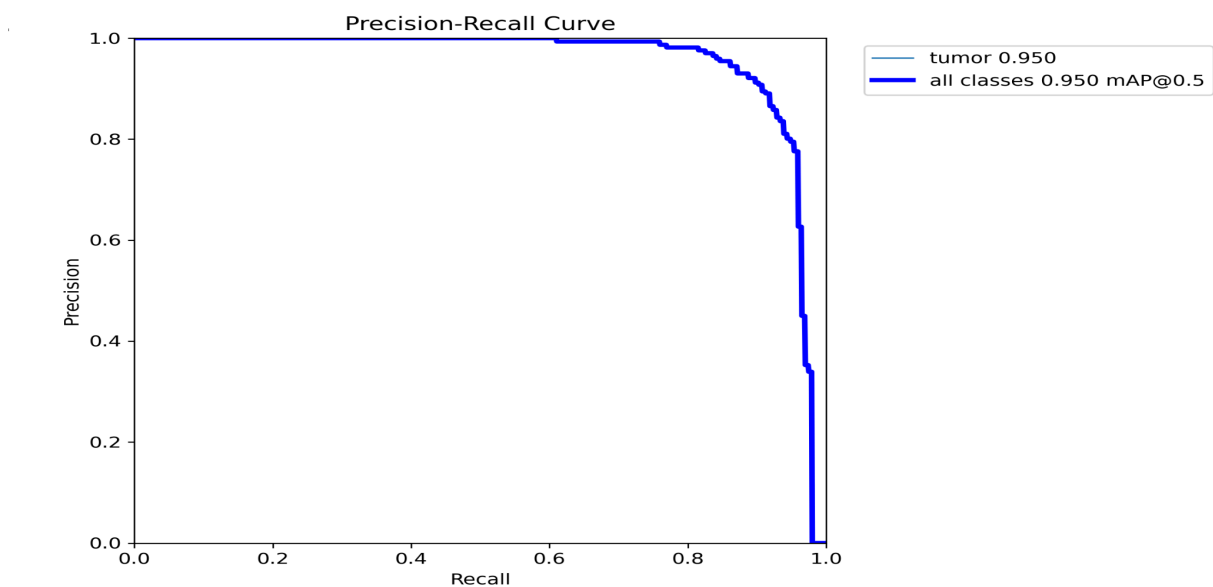
The final evaluation produced the following results:

Metric	Value
Precision	0.91
Recall	0.90
F1 Score	0.91
mAP@0.5	0.95
mAP@0.5:0.95	0.79

These results indicate that the model is capable of detecting brain tumors with high accuracy while maintaining a good balance between precision and recall.

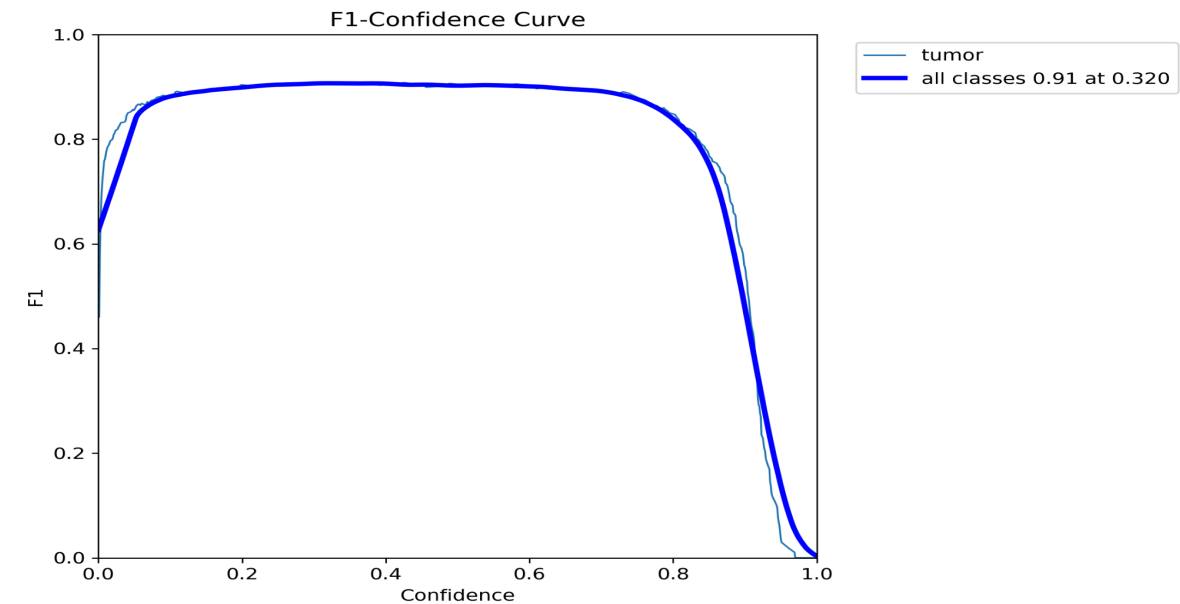
6.3 Precision-Recall Curve

The Precision–Recall curve demonstrates the relationship between precision and recall at different confidence thresholds. The curve remains close to the upper-right corner, indicating strong overall detection performance.



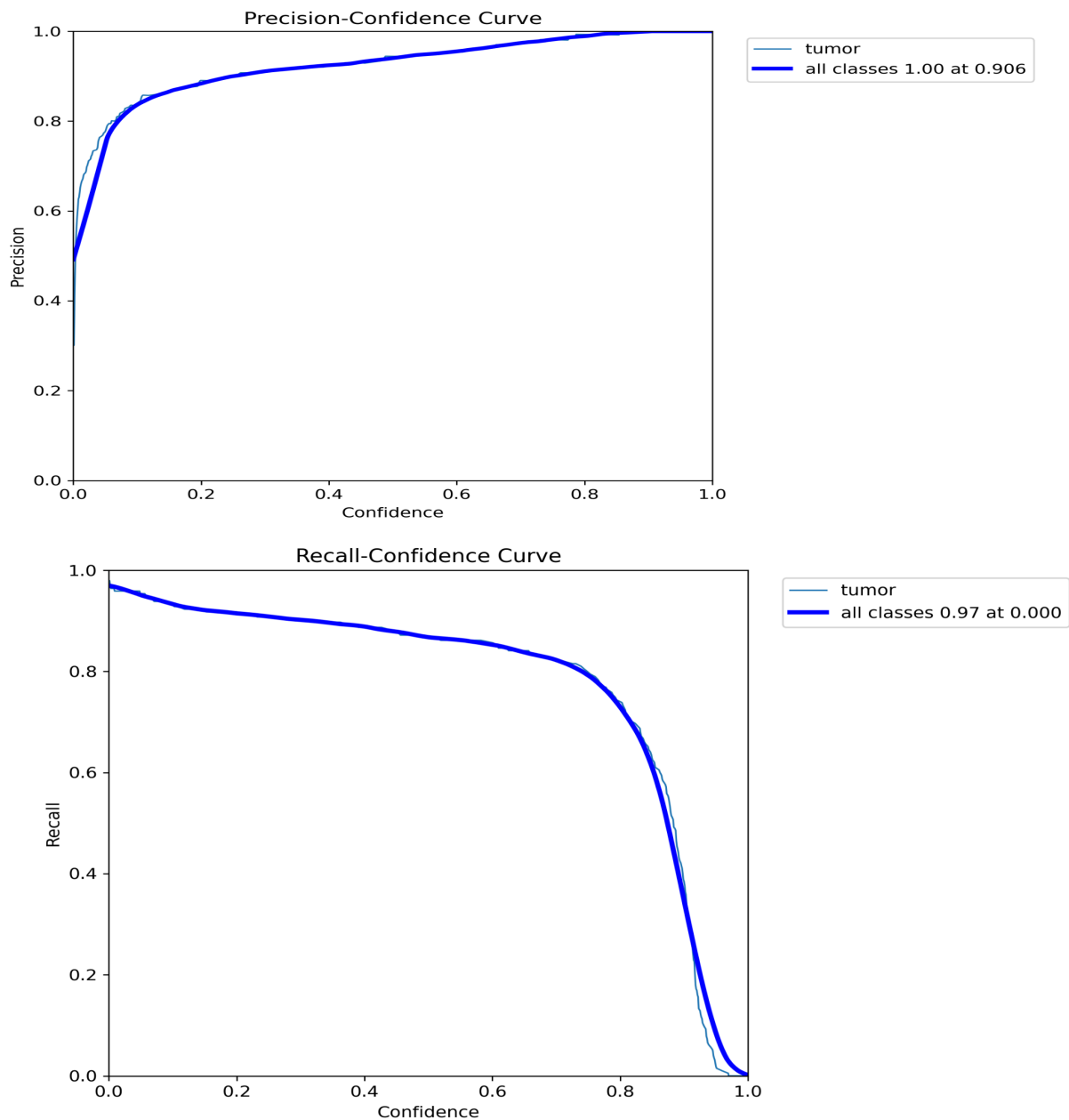
6.4 F1-Confidence Curve

The F1-Confidence curve shows how the F1 score changes with different confidence thresholds. The model achieves its best F1 score of approximately 0.91, suggesting that an intermediate confidence threshold provides the best balance between precision and recall.



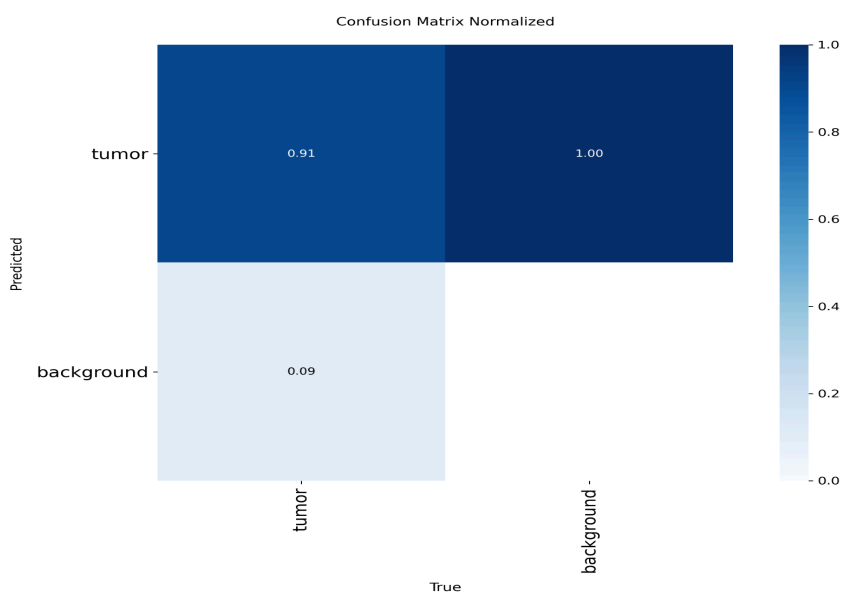
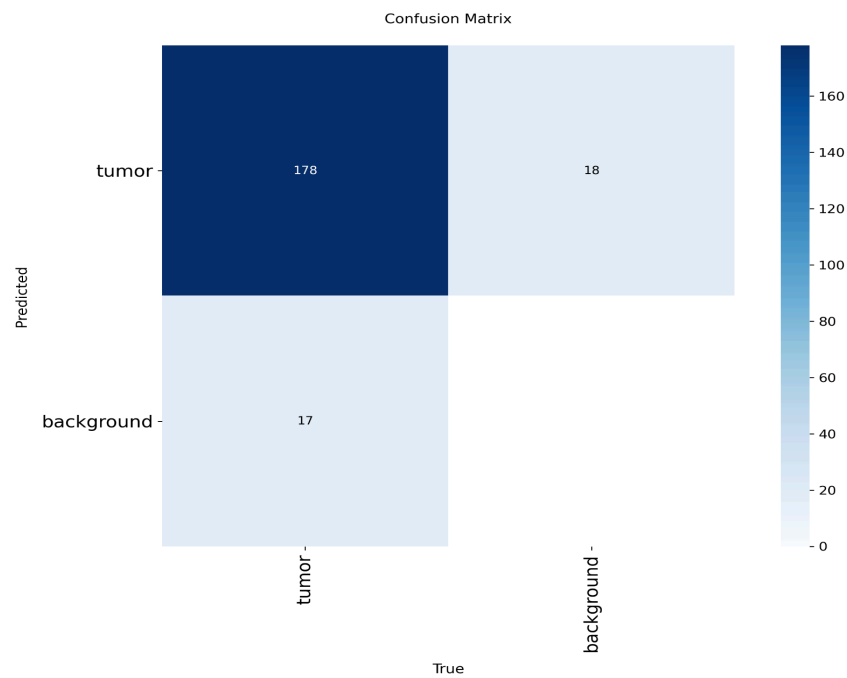
6.5 Precision and Recall Curves

The Precision-Confidence curve shows that precision increases as the confidence threshold becomes higher, while the Recall-Confidence curve shows a gradual decrease in recall at higher confidence values. This behavior is expected because stricter confidence thresholds reduce false positives but may miss some true detections.



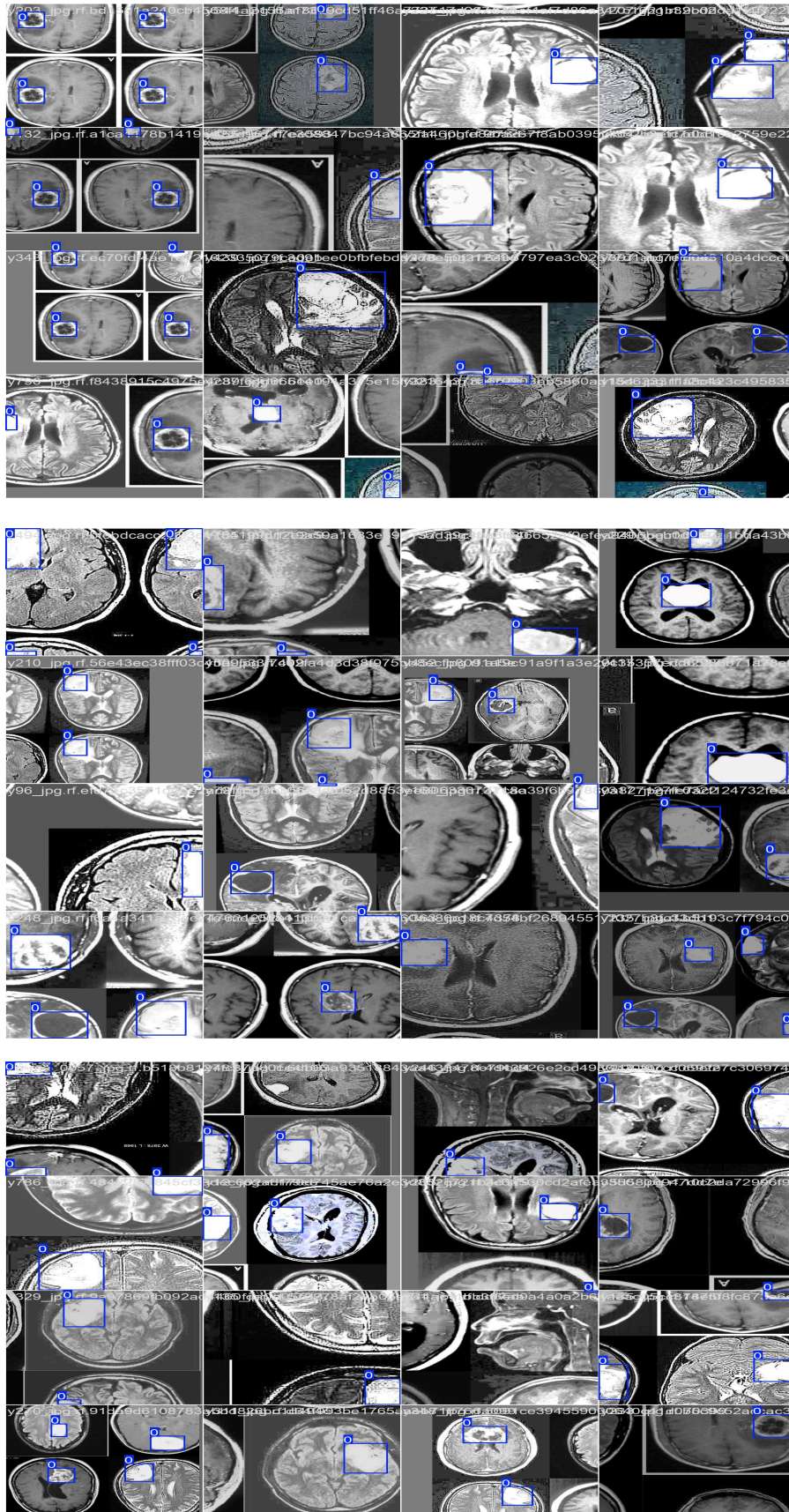
6.6 Confusion Matrix

The confusion matrix shows that the majority of tumor instances were correctly detected by the model. Only a small number of tumors were missed or incorrectly predicted, indicating reliable overall performance.



6.7 Prediction Results

The validation predictions demonstrate that the trained model accurately identifies tumor regions across different MRI scans. Most tumors are enclosed within bounding boxes with high confidence scores, showing good localization capability.



7. Conclusion

In this study, an automated framework for detecting brain tumors was developed by leveraging the YOLOv8 object detection architecture. The model, trained on curated MRI data, successfully identified and localized tumorous regions with high precision.

The evaluation metrics revealed robust performance, achieving an **mAP@0.5 of 0.95**, a **precision of 0.91**, and a **recall of 0.90**. Validation results confirmed that the system accurately generates bounding boxes with high confidence levels. Such findings suggest that YOLOv8 is a viable tool for enhancing the detection of brain tumors in clinical imaging.

While this model is not a replacement for clinical expertise, it serves as a valuable clinical assistant, streamlining preliminary scan analysis and aiding radiologists in identifying critical areas of concern.

8. Limitations

Several constraints were identified during this project:

- The training data was restricted to a single class (tumor).
 - Detection efficacy is tied to the resolution and variety of MRI scans.
 - The system provides localization only, lacking tumor classification or grading capabilities.
 - Evaluation using real-world hospital data has not yet been conducted.
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9. Future Work

Future iterations of this research could focus on:

- Utilizing larger, heterogeneous MRI datasets for training.
 - Adding multi-class support for glioma, meningioma, and pituitary tumor differentiation.
 - Enhancing robustness through advanced hyperparameter optimization and data augmentation.
 - Developing a specialized web or mobile interface for real-time diagnostic support.
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10. References

1. Jocher, G., et al. **Ultralytics YOLOv8 Documentation.**
2. Roboflow Universe. **Brain MRI Tumor Detection Dataset.**
3. Redmon, J., et al. *You Only Look Once: Unified, Real-Time Object Detection.* CVPR, 2016.
4. Current literature on brain tumor detection.